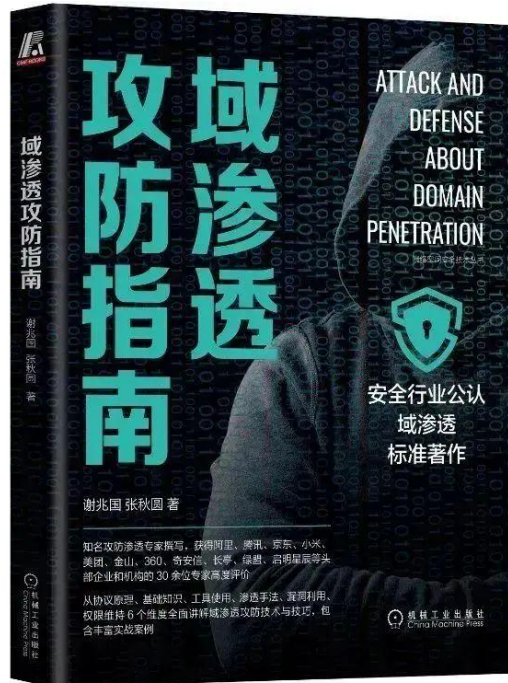


ADEplorer 的使用

本文部分节选于《域渗透攻防指南》，购买请长按如下图片扫码



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ADEplorer

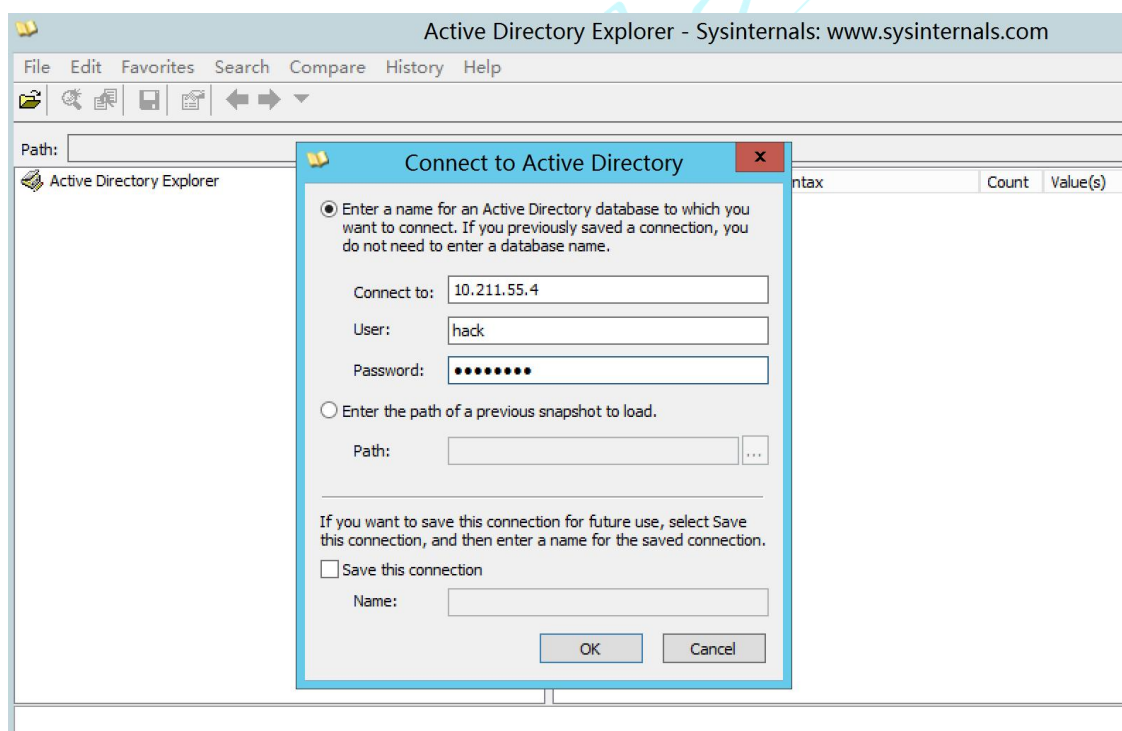
ADEplorer 是域内一款信息查询工具，它是独立的可执行文件，无需安装。它能够列出域组织架构、用户账号、计算机账号等，它可以帮助你寻找特权用户和数据库服务器等敏感目标。下载地址：<http://live.sysinternals.com>

live.sysinternals.com - /

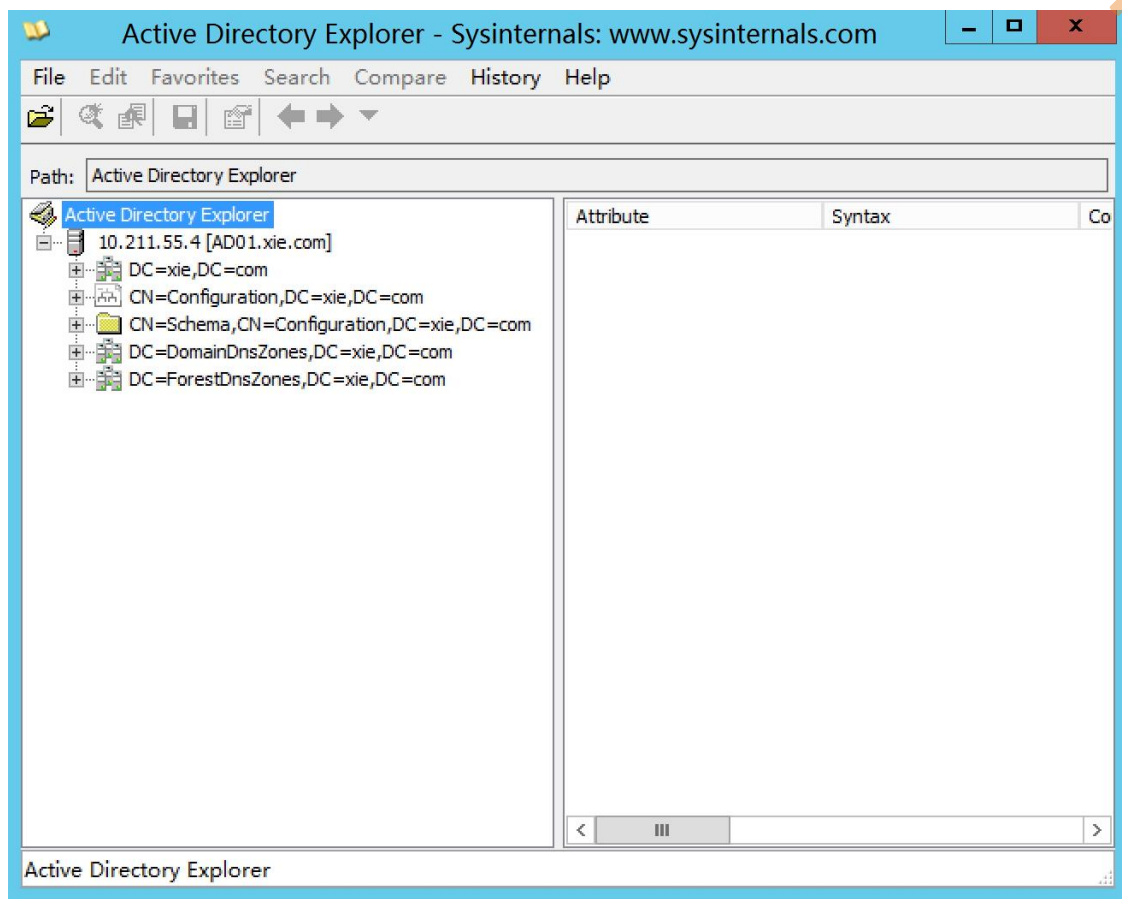
Friday, August 20, 2021 4:54 PM	670	about_this_site.txt
Wednesday, May 11, 2022 5:20 PM	1468320	accesschk.exe
Wednesday, May 11, 2022 5:20 PM	810416	accesschk64.exe
Thursday, September 29, 2022 8:42 PM	264088	AccessEnum.exe
Monday, November 28, 2022 5:53 PM	50379	AdExplorer.chm
Monday, November 28, 2022 5:53 PM	1259968	AdExplorer.exe
Monday, November 28, 2022 5:53 PM	661912	AdExplorer64.exe
Monday, September 14, 2020 2:43 AM	401616	ADInsight.chm
Monday, September 14, 2020 2:36 AM	5106056	ADInsight.exe

连接

下载了 AdExplorer.exe 之后双击打开，会弹出连接的对话框。如图所示，输入域控的 IP 地址，以及提供一个有效的域凭据即可进行连接。

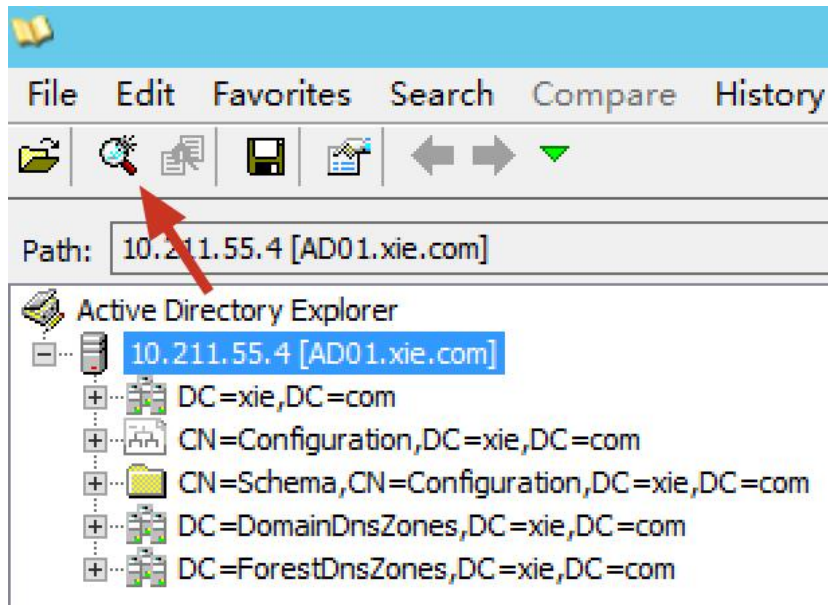


连接成功后如图所示。



查询过滤

点击左上角的放大镜按钮



即可弹出查询对话框，如下我们查询过滤 `objectCategory=computer` 的对象，也就是查询域内的所有机器。可以看到查询出如下对象。

Search Container

Search for objects with the following attributes:

Class: -- Common classes --

Attribute: objectCategory

Relation: is

Value:

(objectCategory=computer)

Add Remove

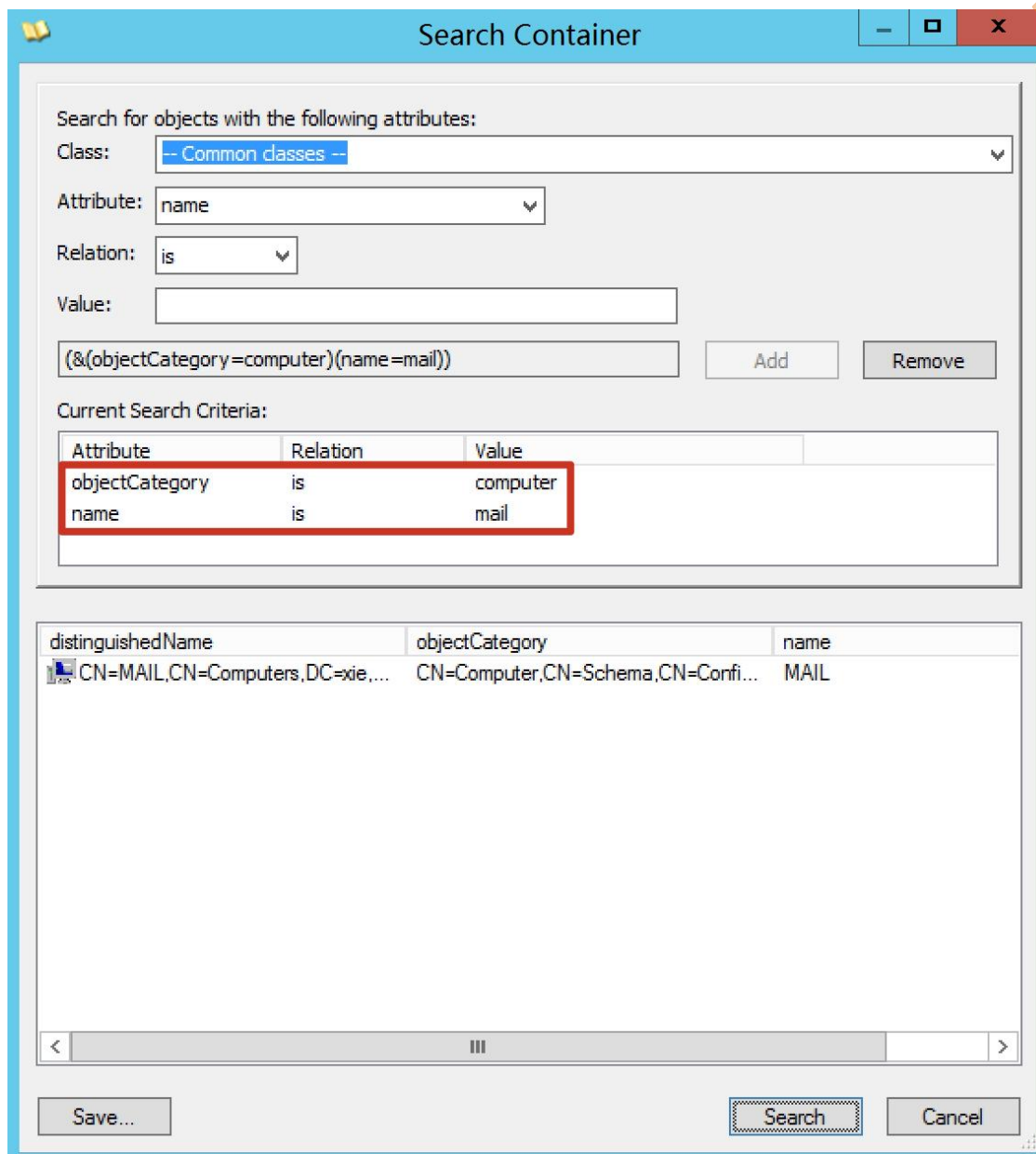
Current Search Criteria:

Attribute	Relation	Value
objectCategory	is	computer

distinguishedName	objectCategory
CN=AD01,OU=Domain Controllers,...	CN=Computer,CN=Schema,CN=Configuration,DC=xie,...
CN=MAIL,CN=Computers,DC=xie,...	CN=Computer,CN=Schema,CN=Configuration,DC=xie,...
CN=WIN10,CN=Computers,DC=xie,...	CN=Computer,CN=Schema,CN=Configuration,DC=xie,...
CN=WIN7,CN=Computers,DC=xie,...	CN=Computer,CN=Schema,CN=Configuration,DC=xie,...

Save... Search Cancel

如果想在增加一个过滤条件 name=mail，也就是查询 mail 机器，操作如下



然后双击查询出的 mail 机器，即可以看到 mail 机器的属性。

Attribute Syntax Count Value(s)

accountExpires	Integer8	1	0x7FFFFFFF
badPasswordTime	Integer8	1	2021/7/1 21:24:06
badPwdCount	Integer	1	0
cn	DirectoryString	1	MAIL
codePage	Integer	1	0
countryCode	Integer	1	0
distinguishedName	DN	1	CN=MAIL,CN=Computers,DC=xie...
dnsHostName	DirectoryString	1	mail.xie.com
dsCorePropagationData	GeneralizedTime	4	2021/12/20 15:23:23;2021/3/7 1...
instanceType	Integer	1	4
isCriticalSystemObject	Boolean	1	FALSE
lastLogoff	Integer8	1	0x0
lastLogon	Integer8	1	2022/2/14 22:57:08
lastLogonTimestamp	Integer8	1	2022/2/14 16:09:22
localPolicyFlags	Integer	1	0
logonCount	Integer	1	175
memberOf	DN	4	CN=Exchange Install Domain Ser...
msDS-SupportedEncryp...	Integer	1	28
msExchCapabilityIdenti...	Integer	1	1
msExchRMSComputerA...	DN	1	CN=FederatedEmail,4c1f4d8b-8...
name	DirectoryString	1	MAIL
ntSecurityDescriptor	NTSecurityDescriptor	1	D:AI(OA;;WP;5f202010-79a5-11...
objectCategory	DN	1	CN=Computer,CN=Schema,CN=...
objectClass	OID	5	top;person;organizationalPerson...
objectGUID	OctetString	1	{C7013A79-8AE0-4EEC-8DF6-22...
objectSid	Sid	1	S-1-5-21-1313979556-36241294...
operatingSystem	DirectoryString	1	Windows Server 2012 R2 Stand...
operatingSystemVersion	DirectoryString	1	6.3 (9600)
primaryGroupID	Integer	1	515
pwdLastSet	Integer8	1	2022/2/14 16:09:33
sAMAccountName	DirectoryString	1	MAIL\$
sAMAccountType	Integer	1	805306369
servicePrincipalName	DirectoryString	24	IMAP/MAIL;IMAP/mail.xie.com;I...
userAccountControl	Integer	1	4096
uSNChanged	Integer8	1	0xD2E3
uSNCreated	Integer8	1	0x5013
whenCreated	GeneralizedTime	1	2022/2/14 16:09:33

Search Container

Search for objects with the following attributes:

Class: Common classes

Attribute: name

Relation: is

Value: (&(objectCategory=computer)(name=mail))

Current Search Criteria:

Attribute	Relation	Value
objectCategory	is	computer
name	is	mail

distinguishedName objectCategory name

CN=MAIL,CN=Computers,DC=xie... CN=Computer,CN=Schema,CN=Corri... MAIL

双击

Save... Search Cancel

增加删除修改属性

如果连接的域账号权限够大的话，还可以增加、删除、修改对象的属性。

如下，选中 hack 用户的任意属性，右键即可看到 Modify(修改)和 Delete(删除)按钮。

Attribute	Syntax	Count	Value(s)
accountExpires	Integer8	1	0x0
badPasswordTime	Integer8	1	2023/1/9 9:33:59
badPwdCount	Integer	1	0
cn	DirectoryString	1	hack
codePage	Integer	1	0
countryCode	Integer	1	0
displayName	DirectoryString	1	hack
distinguishedName	DN	1	CN=hack,CN=Users,DC=xie,DC=com
dSCorePr	Integer	2	2021/12/20 15:23:23;1601/1/1 7:00:01
homeMDB	Integer	1	CN=Mailbox Database 0766714717,CN:
instanceT	Integer	1	4
lastLogof	Integer	1	0x0
lastLogon	Integer	1	2023/1/9 9:34:11
lastLogon	Integer	1	2023/1/5 22:19:05
legacyEx	Integer	1	/o=First Organization/ou=Exchange Ad
logonCount	Integer	1	19
logonHours	OctetString	1	255 255 255 255 255 255 255 255
mail	DirectoryString	1	hack@xie.com

Properties...

Copy Attributes

Display Integers as ▸

Modify...

Delete

New Attribute...

如果想增加属性，点击 New Attribute

distinguishedName	DN	1
dSCorePr	Integer	2
homeMDB	Integer	1
instanceT	Integer	1
lastLogof	Integer	1
lastLogon	Integer	1
lastLogon	Integer	1
legacyEx	Integer	1
logonCount	Integer	1
logonHours	OctetString	1

Properties...

Copy Attributes

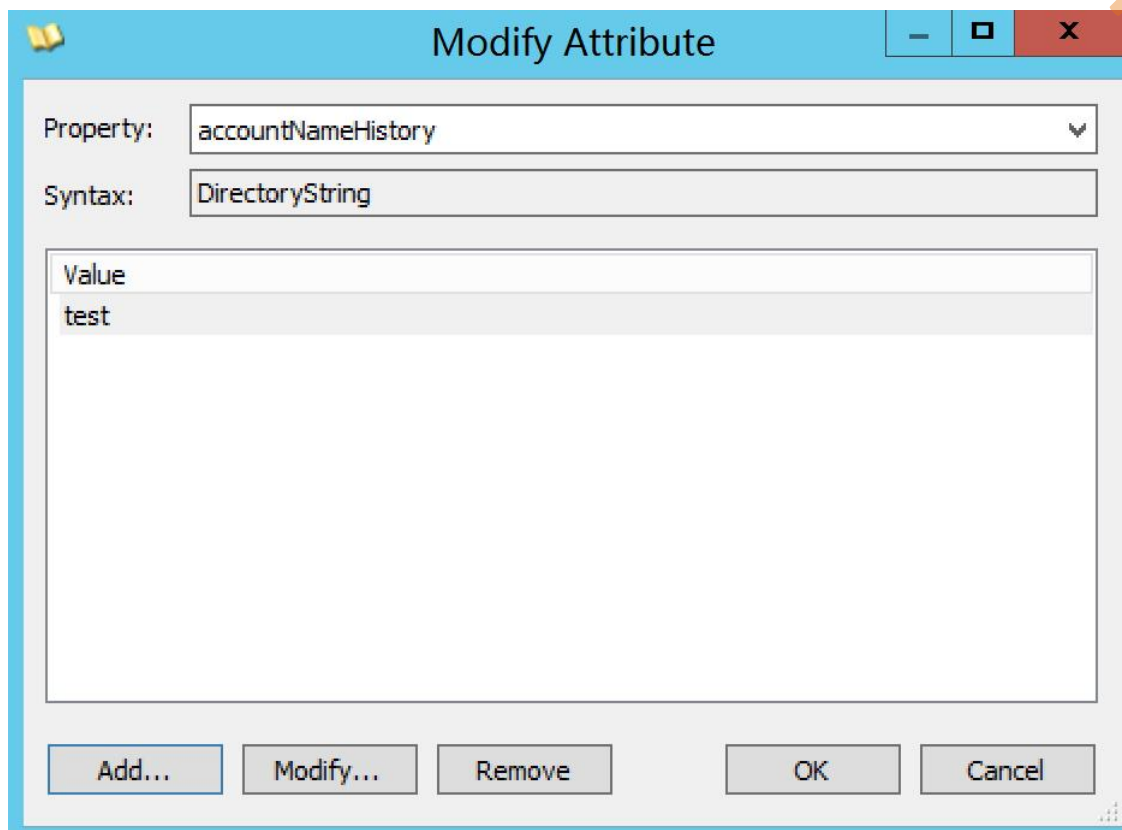
Display Integers as ▸

Modify...

Delete

New Attribute...

然后在弹出的对话框中选择要添加的属性，以及输入对应的 Value 即可。这里新增的属性为 accountNameHistory，新增的值为 test。

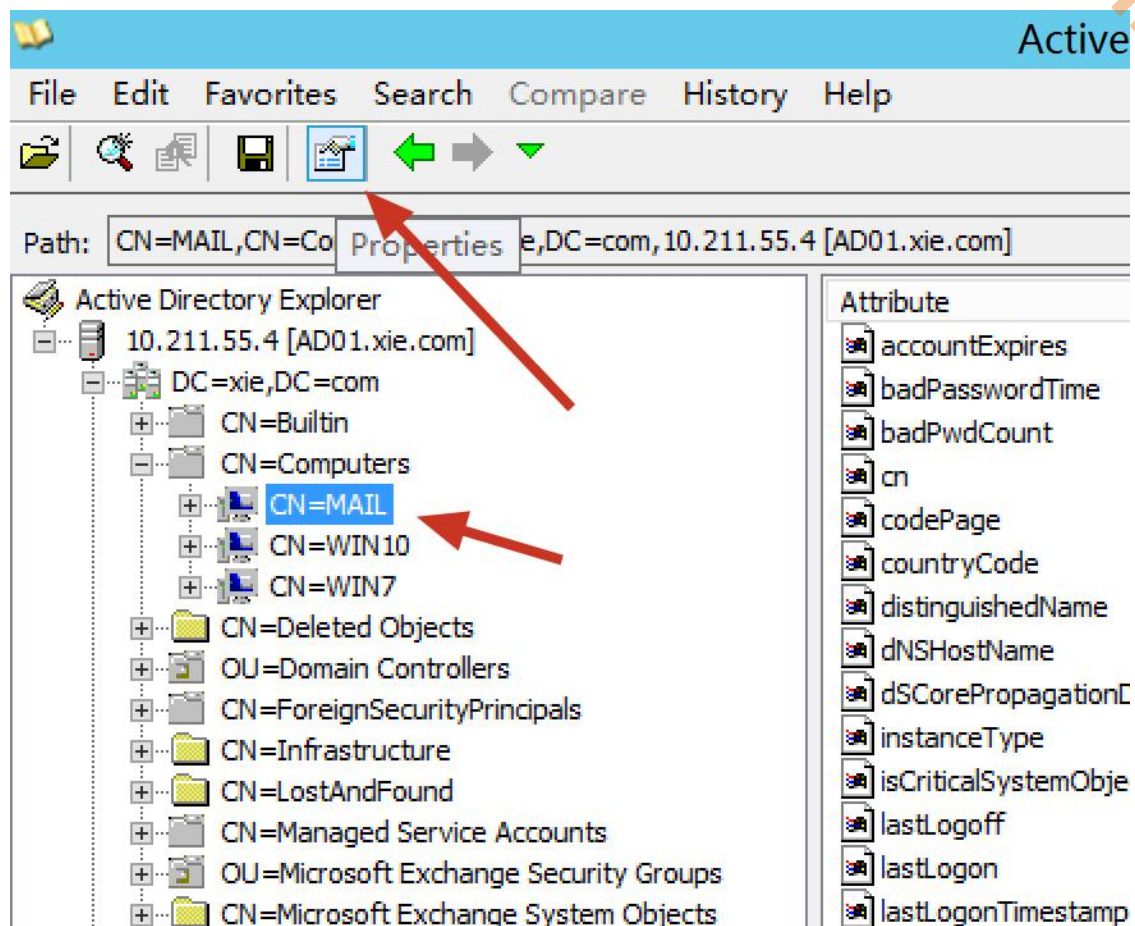


即可看到新增的属性。

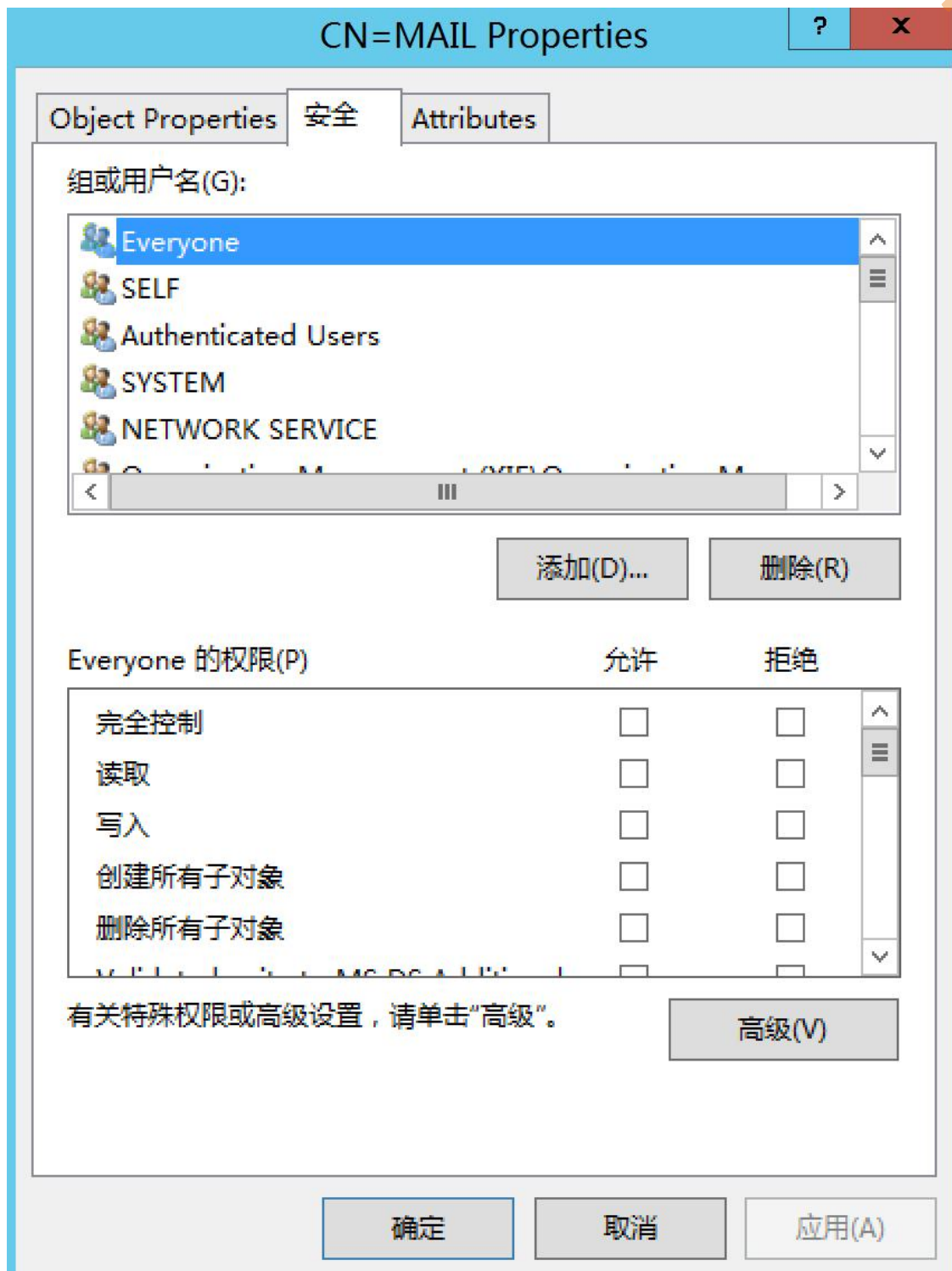
Attribute	Syntax	Count	Value(s)
accountExpires	Integer8	1	0x0
accountNameHistory	DirectoryString	1	test
badPasswordTime	Integer8	1	2023/1/9 9:33:59
badPwdCount	Integer	1	0
cn	DirectoryString	1	hack
codePage	Integer	1	0
countryCode	Integer	1	0
displayName	DirectoryString	1	hack
distinguishedName	DN	1	CN=hack,CN=Users,DC
dSCorePropagationData	GeneralizedTime	2	2021/12/20 15:23:23;1

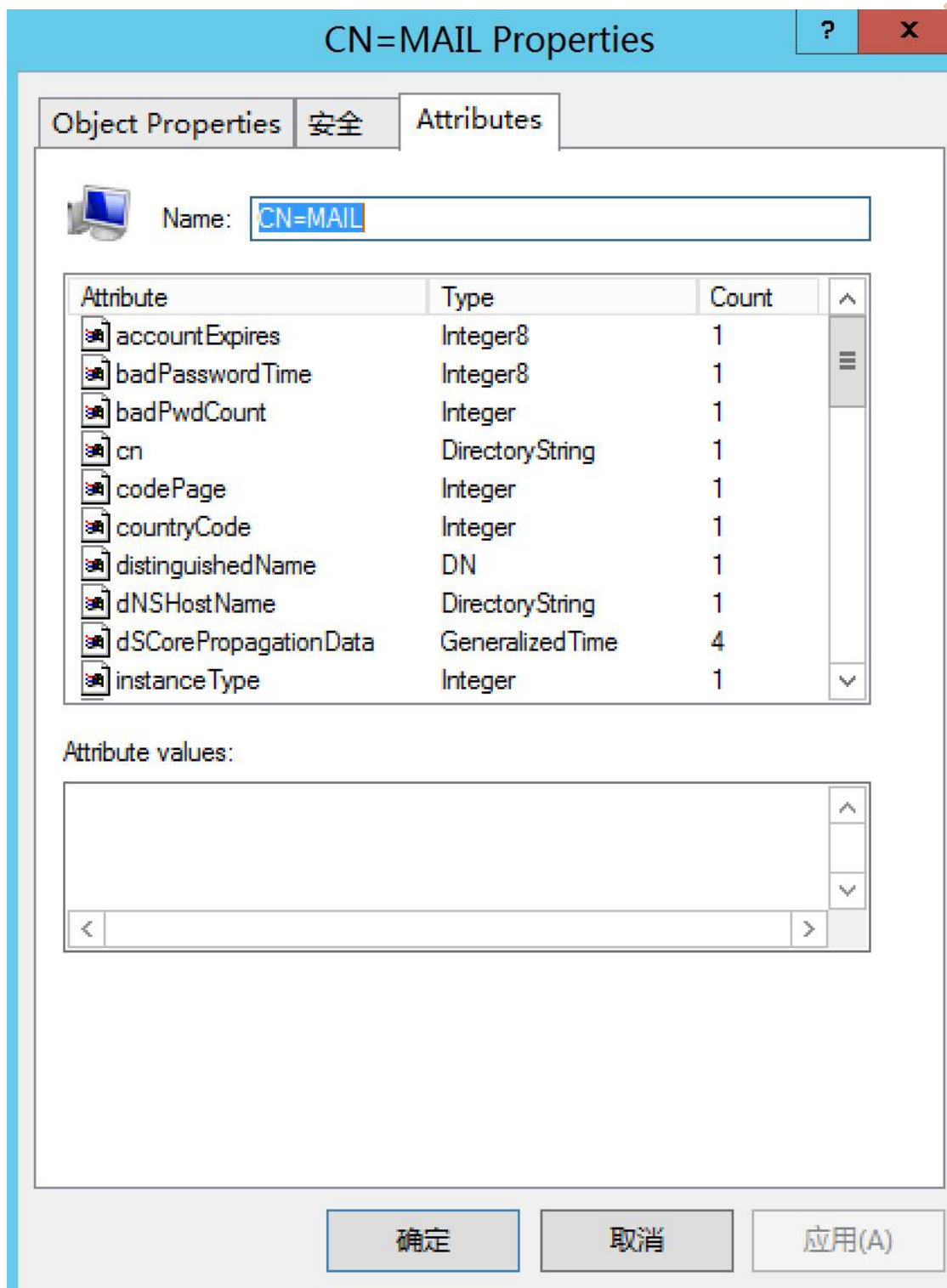
查看对象属性

选中任意一个对象，然后点击左上角的如图按钮



即可看到对象的属性





也可以直接在右边查看到对象的属性

Active Directory Explorer - Sysinternals: www.sysinternals.com [10.211.55.4 [AD01.xie.com]]

Path: CN=MAIL,CN=Computers,DC=xie,DC=com,10.211.55.4 [AD01.xie.com]

Attribute	Syntax	Count	Value(s)
accountExpires	Integer8	1	0x7FFFFFFF
badPasswordTime	Integer8	1	2021/7/1 21:24:06
badPwdCount	Integer	1	0
cn	DirectoryString	1	MAIL
codePage	Integer	1	0
countryCode	Integer	1	0
distinguishedName	DN	1	CN=MAIL,CN=Computers,DC=xie,DC=com
dnsHostName	DirectoryString	1	mail.xie.com
dsCorePropagationData	GeneralizedTime	4	2021/12/20 15:23:23;2021/3/7 17:28:31;2021/3/7 17:28:31;1601/1/2 1:16:33
instanceType	Integer	1	4
isCriticalSystemObject	Boolean	1	FALSE
lastLogoff	Integer8	1	0x0
lastLogon	Integer8	1	2022/2/14 22:57:08
lastLogonTimestamp	Integer8	1	2022/2/14 16:09:22
localPolicyFlags	Integer	1	0
logonCount	Integer	1	175
memberOf	DN	4	CN=Exchange Install Domain Servers,CN=Microsoft Exchange System Objects,DC=xie,DC=com;CN=Managed Availability Servers,OU=Microsoft Exchange Sec
msDS-SupportedEncryp...	Integer	1	28
msExchCapabilityIdent...	Integer	1	1
msExchRMSCComputerA...	DN	1	CN=FederatedEmail-4c1f468b-8179-4148-930f-00a95fa1e042,CN=Users,DC=xie,DC=com
name	DirectoryString	1	MAIL
ntSecurityDescriptor	NTSecurityDescriptor	1	D:AI(OA;);WP;S(02010-79a5-11d0-9020-00c04fc2d4cfbf967a86-0d66-11d0-a285-00a003049e2dA)(OA;);WP;bf967950-0d66-11d0-a285-00a003049e2dA
objectCategory	DN	1	CN=Computer,CN=Schema,CN=Configuration,DC=xie,DC=com
objectClass	OID	5	top;person;organizationalPerson;user;computer
objectGUID	OctetString	1	{C7013A79-8AE0-4EEC-80F6-224690F36AE0}
objectId	Sid	1	S-1-5-21-1313979556-3624129433-4055499191-1107
operatingSystem	DirectoryString	1	Windows Server 2012 R2 Standard
operatingSystemVersion	DirectoryString	1	6.3 (9600)
primaryGroupID	Integer	1	515
pwdLastSet	Integer8	1	2022/2/14 16:09:33
sAMAccountName	DirectoryString	1	MAIL\$
sAMAccountType	Integer	1	805336369
servicePrincipalName	DirectoryString	24	IMAP/MAIL;IMAP/mail.xie.com;IMAP4/MAIL;IMAP4/mail.xie.com;POP/MAIL;POP/mail.xie.com;POP3/MAIL;POP3/mail.xie.com;exchange6FR/MAIL;exchange6FR
userAccountControl	Integer	1	4096
uSNChanged	Integer8	1	0x02E3
uSNCreated	Integer8	1	0x5013
whenChanged	GeneralizedTime	1	2022/2/14 16:09:33
whenCreated	GeneralizedTime	1	2021/3/7 16:06:18

拍摄快照

域内机器

域内机器可以直接利用 ADEplorer 执行如下命令将域信息导出成.dat 文件格式

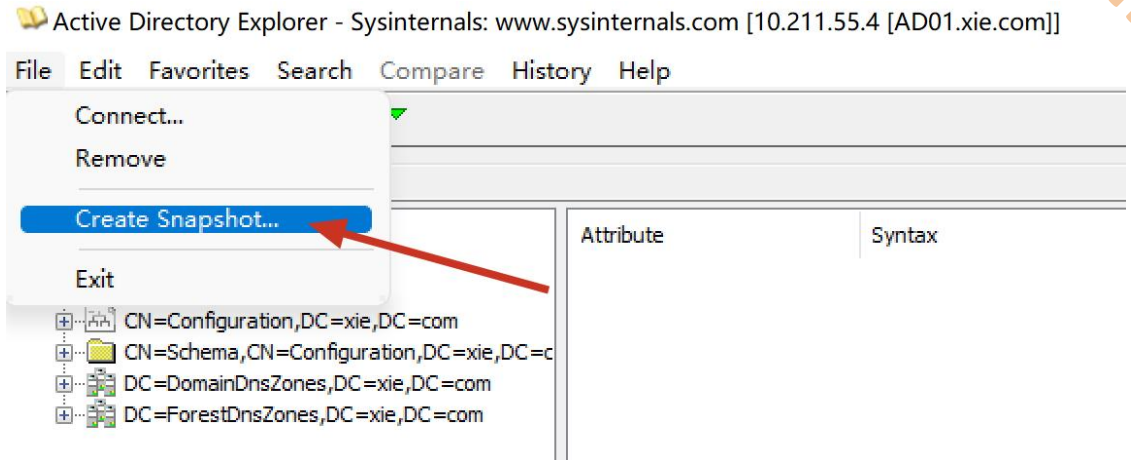
ADEplorer.exe -snapshot "" result.dat /accepteula

```
C:\Users\Administrator\Desktop>ADEplorer.exe -snapshot "" result.dat /accepteula

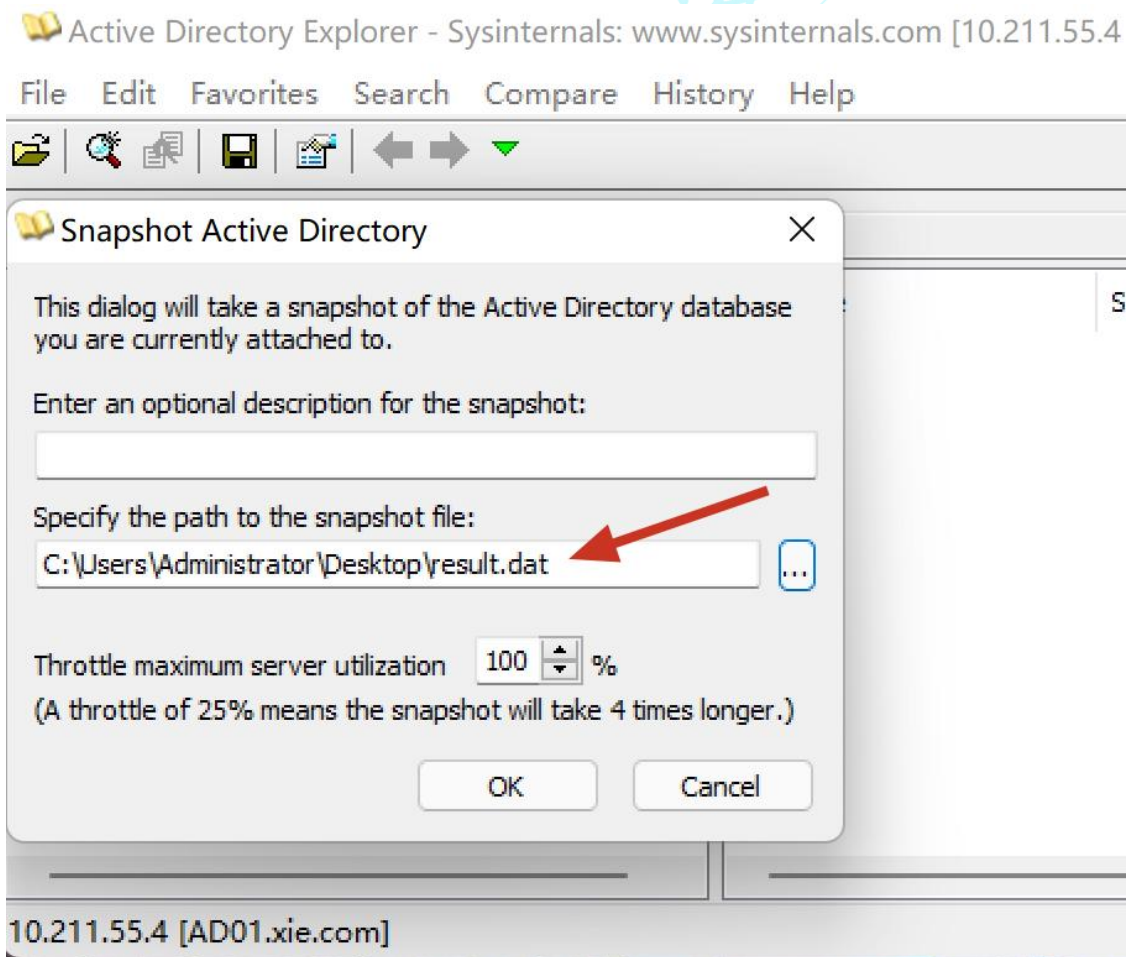
C:\Users\Administrator\Desktop>dir | findstr result.dat
2022/04/14 20:31          12,582,912 result.dat
```

非域内机器

首先，使用 ADEplorer 利用有效账号密码连接 ldap。然后点击 File——>Create Snapshot



如下输入保存路径即可。

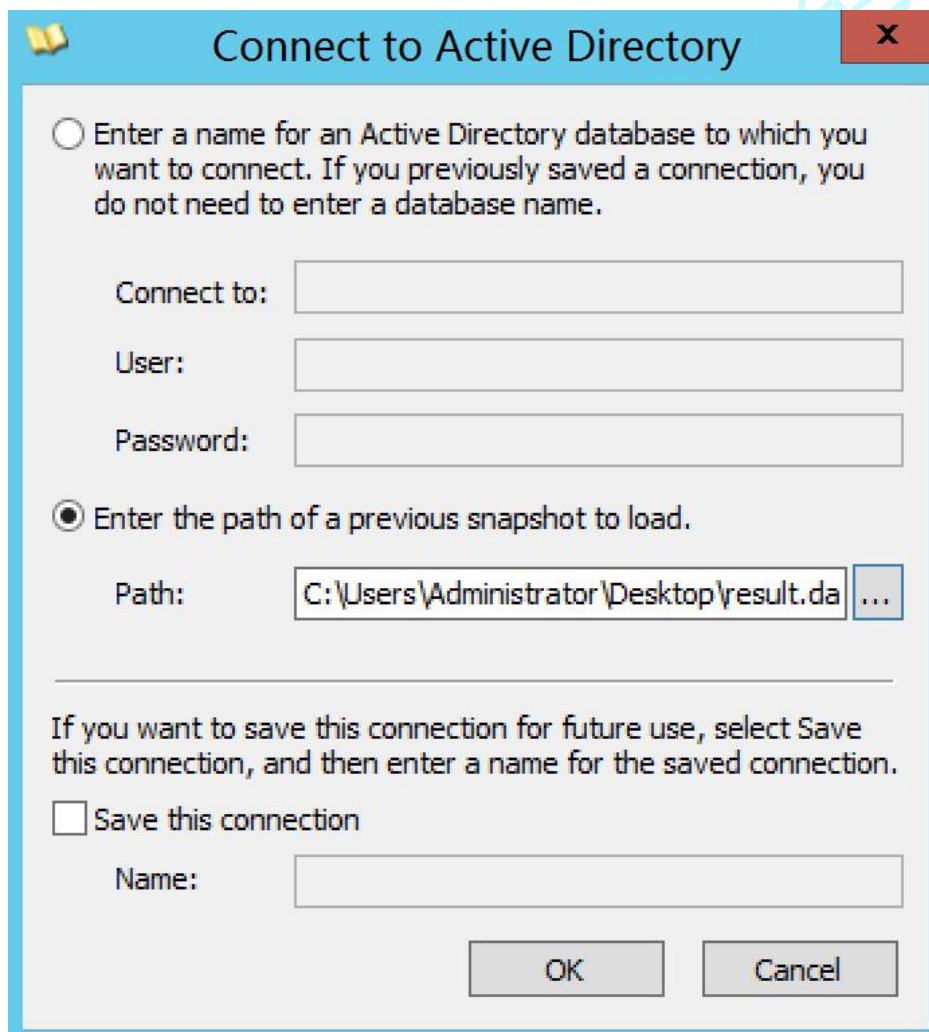


注：由于不同权限的用户查看的 LDAP 信息不同，因此不同权限的用户拍摄的快照内容也不同。

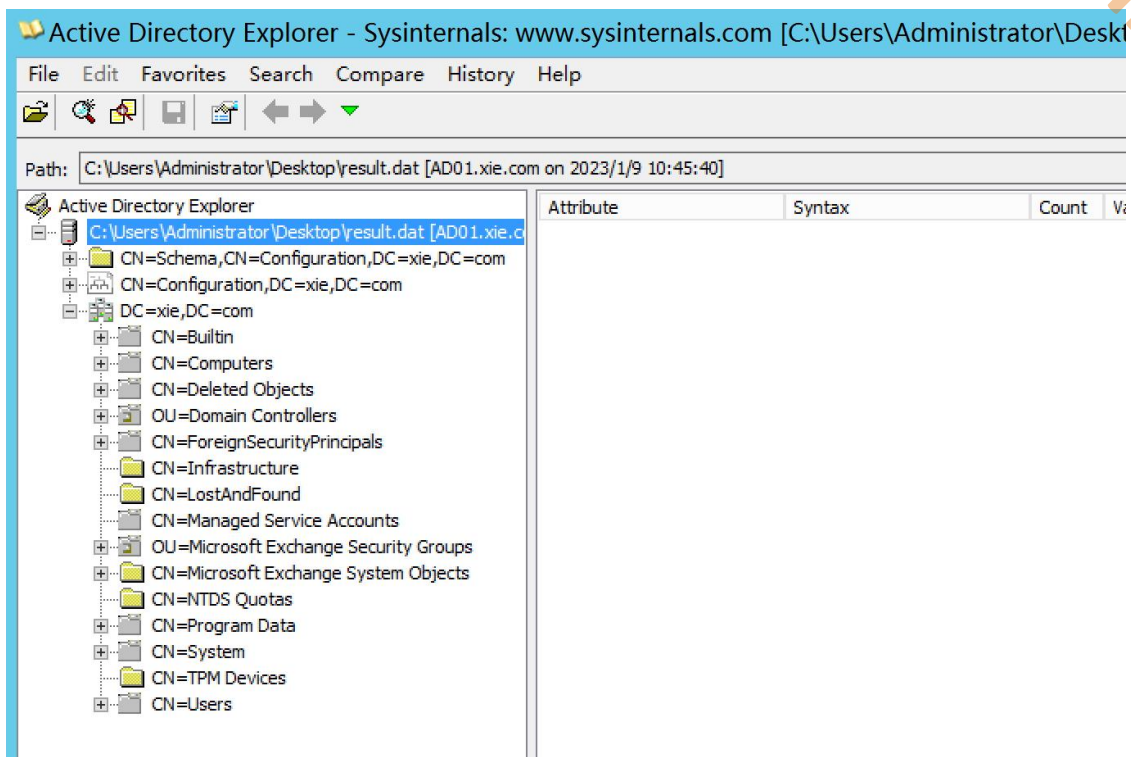
导入快照

对于导出的快照，我们可以本地直接导入快照，本地导入快照不会改变目标与域的任何设置，非常适合用于进行信息收集。

双击打开 ADEplorer.exe，然后如图操作 result.dat 快照。



即可看到快照已经导入。



使用 ADEplorerSnapshot.py 解析快照

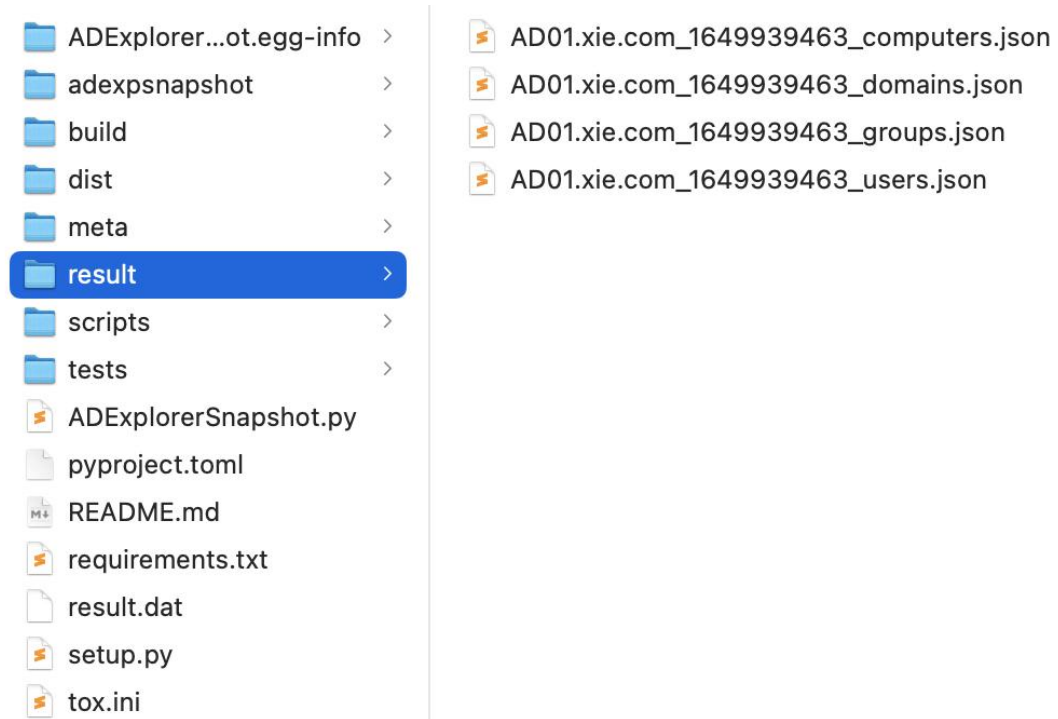
也可以使用 ADEplorerSnapshot.py 将 dat 文件解析为 BloodHound 可解析的格式。

项目地址: <https://github.com/c3c/ADEplorerSnapshot.py>

python3 ADEplorerSnapshot.py result.dat -o result

```
→ ADEplorerSnapshot python3 ADEplorerSnapshot.py result.dat -o result
[*] Server: AD01.xie.com
[*] Time of snapshot: 2022-04-14T20:31:03
[*] Mapping offset: 0x298cac
[*] Object count: 3801
[+] Parsing properties: 1473
[+] Parsing classes: 264
[+] Parsing object offsets: 3801
[+] Preprocessing objects: 93 sids, 4 computers, 1 domains with 2 DCs
[+] Collecting data: 8 users, 46 groups, 4 computers, 0 trusts
[+] Output written to AD01.xie.com_1649939463_*.json files
```

生成如下格式的文件



导入 BloodHound

最后将生成的 json 文件直接拖入到 BloodHound 中即可。

Upload Progress

AD01.xie.com_1649939463_computers.json

Upload Complete 100%

AD01.xie.com_1649939463_domains.json

Upload Complete 100%

AD01.xie.com_1649939463_groups.json

Upload Complete 100%

Clear Finished





Search for a node



Database Info

Node Info

Analysis

DB STATS

Address	bolt://localhost:7687
DB User	neo4j
Sessions	0
Relationships	508
ACLs	463
Azure Relationships	0

ON-PREM OBJECTS

Users	9
Groups	50
Computers	4
OUS	0
GPOs	0
Domains	2

注：使用 ADEplorerSnapshot.py 解析的文件只能使用 BloodHound>=4.1 版本才能打开。

非常感谢您读到现在，由于作者的水平有限，编写时间仓促，文章中难免会出现一些错误或者描述不准确的地方，恳请各位师傅们批评指正。如果你想一起学习 AD 域安全攻防的话，可以加入下面的知识星球一起学习交流。



知识星球卡片，背景为橙红色。顶部有谢公子的头像和名字。中间部分展示了书籍封面《域渗透攻防指南》及其星主信息。下方统计了成员数量、内容数量和运营天数。底部有书籍简介、现价¥199、微信扫码加入星球的提示以及一个二维码。底部导航栏包含“知识星球”和“谢公子学安全”的标识。

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